Andrew Baughns

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Mobile Developer | AI Specialist

Technically astute and solution-oriented professional specializing in mobile development, AI-powered automation, and on-device machine learning. Adept at integrating AI into real-world applications, enhancing user experiences, and automating workflows using cutting-edge LLMs and NLP technologies.

Track record of leveraging cutting-edge technologies and development practices, including reactive programming, modular design, and CI/CD pipelines to deliver high-performing, user-centric applications. Proficient in modern Android frameworks, machine learning integration, and AI technologies. Proven expertise in leading cross-functional teams, streamlining SDLC processes, and executing test-driven development to deliver high-quality, maintainable solutions. Out-of-box thinker with strong communication, analytical problem-solving.

Areas of Expertise

- Mobile Application Development
- Bluetooth & IoT Systems
- User Interface Modernization
- Software Development Life Cycle
- Analytical Problem-Solving
- Large Language Models
- AI-Powered Analytics
- On-Device AI Deployment
- Application Crash Analytics
- Privacy-First Development Practices
- User-Centered Design Practices
- Software Reliability Enhancements
- API Integration & Development
- Scalable Software Architecture
- Cross-functional Collaboration

Technical Proficiencies

Programming Languages:	Java | Kotlin | Python | C# | SQL | JavaScript | HTML | MATLAB | VBScript | Shell Scripting | Bash
Machine Learning & AI Tools:	Large Language Models (LLM) | Natural Language Processing (NLP) | Deep Learning | Evolutionary Algorithms | Convolutional Neural Networks (CNN) | Vision Transformers
Frameworks & Libraries:	Android Framework | React.js | RxJava | PyTorch | TensorFlow | TensorFlowLite | Scikit-Learn | NumPy | Pandas
Cloud Services & Platforms:	AWS EC2 | AWS Lambda | AWS DynamoDB | AWS S3 | AWS API Gateway | Bitrise CI/CD | Firebase | AWS SageMaker | Amazon CloudWatch
Databases:	Microsoft SQL Server | SQLite | NoSQL | AWS DynamoDB
DevOps & CI/CD:	Jenkins | Docker | Bitrise | Continuous Integration (CI)
Visualization & UI/UX Tools:	Jupyter | Figma | Google Docs | Visio | D3
Technologies:	Bluetooth Low Energy | Android Automotive | Workflow Engines | REST APIs

Professional Experience

Cambridge Mobile Telematics – Cambridge, MA **2017 – 2024**
Senior Software Engineer III

Led development, enhancement, and maintenance of multiple Android applications, including integrating new features, improving app performance, and managing app deployment on Play Store. Delivered QA automation and integration testing solutions by creating test cases and developing app automation scripts. Championed productivity-enhancing practices and reduced cognitive load for debugging and feature development by introducing improved documentation and coding standards.

Notable Contributions/Accomplishments:

- Delivered a successful pilot Android app **DriveWell Sg** for **Amazon Flex** delivery drivers in Singapore. Integrated advanced geolocation tracking feature to monitor driver movements while balancing battery life and privacy requirements.

- Directed all phases of software development, including API integration with cloud team, production support documentation, and deployment coordination with product management to the Play Store.
- Enhanced Play Store ratings of **Drivescape smart camera app for Uber** from **3** stars to **4.5+** by introducing innovative features, including video streaming and downloading from dashcams, alongside bug fixes and stability enhancements.
- Designed and built original app prototype and contributed significant updates upon rejoining project in December 2023.
- Advocated test-driven development and modular design to enhance separation of platform interface code from business logic.
- Improved **Android Mobile Sensing SDK** and enabled automated crash analysis and outlier detection via structured application logging. Assisted production support and QA teams in resolving issues by managing production documentation.
- Led technical design and development of family sharing feature for **DriveWell Go app** mobile telematics platform which enabled gamification of safe driving behaviors and tracking of family member. Collaborated on defining communication protocols and APIs to enforce business rules around group management and permissions.
- Modernized Android DriveWell Whitelabel platform and Mobile Sensing SDK by introducing RxJava and reactive programming patterns, enhancing code readability and maintainability while aligning with contemporary development methodologies.
- Established and configured continuous integration for **Android DriveWell Whitelabel** framework and **Android Mobile Sensing SDK** using Bitrise CI/CD tools.

Nuance Communications – Burlington, MA **2012 – 2017**
Senior Software Engineer

Collaborate with cross-functional teams to design, develop, and enhance software applications. Drive innovation by implementing new features, optimizing existing systems, and ensuring alignment with industry best practices. Contribute to development process by ensuring clean code, efficient workflows, and effective testing. Improve user experiences by modernizing interfaces, integrating APIs, and addressing technical challenges proactively.

Notable Contributions/Accomplishments:

Dragon Mobile Assistant (DMA)

- Developed and modernized **Dragon Mobile Assistant (DMA)** Android application, an Android-based **alternative to Siri** by collaborating with a team of eight software engineers.
- Introduced innovative features to the app, such as voice-based texting, alarm setup, and a redesigned browser interface.
- Designed custom screen widgets and wrote database integration and AlarmManager code to elevate user experience.
- Improved user experience by revamping DMA's integrated web browser and incorporating gesture-based navigation for browser tabs and smart thumbnail caching for previews.
- Enabled voice-driven stock portfolio monitoring by integrating Yahoo Finance API with DMA; optimized both client UI and backend functionality to enhance app's service offerings.
- Reduced costs by repurposing existing bespoke continuous integration server on Mac for improved resource efficiency.

Voice-Controlled Car Head Unit App for Ford

- Played a pivotal role in modernizing user interface of Ford's voice-controlled car head unit application.
- Enabled faster feature development and scalability by leading transition to React.js from legacy JQuery and HTML/JavaScript.
- Ported Windows-based embedded client/server web application to ARM Linux, ensuring compatibility with modern hardware.
- Developed a Bing Maps-based simulator to test route guidance by replicating car movements across latitude/longitude coordinates.

Homesite Insurance – Boston, MA **2006 – 2011**
Software Engineer / Team Lead

Cultivated proficiency in SQL and Microsoft SQL Server toolchains as a Junior Developer at Homesite. Created table-driven legal documents for client communications and developed expertise in large-scale customer data modeling, troubleshooting production data, and coordinating with enterprise client-server applications and batch processes. Oversaw all phases of production print-related operations. Managed enterprise document input and output processing across web services, backend batch processes, and fax systems while serving as primary subject matter expert.

Notable Contributions/Accomplishments:

- Modernized document generation process by creating a table-driven workflow engine using C#, .NET Framework, and ADO.NET.
- Enhanced fault tolerance, error monitoring, and recovery capabilities and took full ownership of the system to implement improvements that increased reliability and reduced maintenance challenges.
- Led a team of developers to create a prototype Android home insurance application. Enabled users to access and request policy documents through a web service built on workflow engine infrastructure and introduced natural language document search features to reduce customer service inquiries.

Additional Experience *as* ***Scientific Programmer*** *at Harvard Medical School – Cambridge, MA*

Education

Master of Computer Science, Major: Computer Science **| Specialization:** Interactive intelligence
Georgia Tech – Atlanta, GA *(Anticipated Graduation in 2026)*

Bachelor of Science, Major: Mathematics **| Minor:** International Relations
Massachusetts Institute of Technology – Cambridge, MA

Projects

Personal ChatGPT Server — 2025 – Present

- Built a cloud-based LLM service using AWS EC2, Lambda, and DynamoDB to provide a private, scalable conversational AI backend with support for Llama 70B and DeepSeek r1 models.

AI-Powered Desktop Automation Assistant — 2024 – Present

- Designed an AI-powered desktop automation assistant that enables seamless collaboration between users and LLMs for software architecture and brainstorming.
- Executed shell utility integration, allowing the AI to execute desktop actions under user supervision, automating workflow processes.
- Developed a Python library for LLM dialog management to support prompt engineering, task automation, and integration with OpenAI GPT-4 and LLaMA-based models.
- Demonstrated the project in real-time, showing how the LLM collaborates with users to build a mindmap and perform automated tasks on the desktop.
  [Watch Demo: YouTube] *https://www.youtube.com/watch?v=HINGICAPsCg*
  [GitHub Repository] *https://github.com/hexagonc/local-llm-dialogs*

Stock Trading Bot — 2023

- Developed a Python-based tool for stock data cleaning, back-testing trading strategies, and maximizing expected portfolio gains using technical and fundamental analysis.

Wildlife Tracking with On-Device Machine Vision — 2024

- Implemented on-device machine vision using TensorFlowLite and MediaPipe for wildlife classification, deploying models like ResNet50 and Mobile Vision Transformers.

Android Prototyping App — 2016 – Present

- Designed and engineered educational Android app with Lisp-like scripting language for on-device app prototyping and user interface testing, supporting robotics and speech-controlled systems. The linked Youtube video demonstrates the use of the prototyping app to teach basic programming concepts including an easy introduction to LLMs, prompt engineering and embeddings to teach deep learning concepts to children. In addition, we see the dynamic nature of the user interface, with the user being able to alter app behavior in arbitrary ways to suit one's own tastes.
- ***[Watch Demo: YouTube]*** *https://www.youtube.com/watch?v=fxZvvq2F0Qw&t=1s*